Report (Content)

Predicting Product Review Sentiment for TrendCart Ltd Using Machine Learning

1. Introduction

TrendCart Ltd is a UK-based e-commerce startup that sells lifestyle products through its online platform. As the company grows, it receives a large number of customer reviews. Manually analysing these reviews is time-consuming and inefficient. The objective of this project is to develop a simple machine learning model that can automatically classify customer reviews into Positive, Neutral, and Negative sentiment categories.

2. Dataset Description

The dataset used for this project contains customer product reviews and ratings collected from an e-commerce platform. Two columns were selected for analysis:

- Review Text
- Review Rating (1–5 Stars)

Sentiment labels were generated using the following rules:

- Rating 4–5 → Positive
- Rating 3 → Neutral
- Rating 1–2 → Negative

Missing values were removed before analysis.

3. Data Preprocessing

The following preprocessing steps were performed:

- Selected relevant columns from the dataset.
- Removed null or missing records.
- Created sentiment labels from review ratings.
- Converted review text into numerical features using TF-IDF Vectorization.
- Split the dataset into training and testing sets using an 80:20 ratio.

4. Exploratory Data Analysis (EDA)

Initial analysis showed that most customer reviews were positive. Sentiment distribution was visualized using pie charts and bar graphs. These visualizations helped identify the overall customer satisfaction level.

Key observations:

- Positive reviews formed the majority of the dataset.
- Neutral reviews represented a smaller percentage.
- Negative reviews highlighted areas where customer experience could be improved.

5. Machine Learning Model

A Logistic Regression classifier was used because it is simple, efficient, and performs well on text classification tasks.

Feature Extraction Technique:

- TF-IDF Vectorizer

Classification Algorithm:

- Logistic Regression

The model was trained on the training dataset and evaluated using unseen test data.

6. Model Performance

Performance was measured using Accuracy Score and Classification Report.

Results indicated that the model successfully classified customer reviews with a high level of accuracy. The model demonstrated its ability to identify positive, neutral, and negative sentiments from review text.

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

7. Business Relevance

This solution provides TrendCart with a low-cost and practical method for analysing customer feedback.

Benefits include:

- Faster review monitoring.
- Identification of customer satisfaction trends.
- Early detection of product-related issues.
- Support for marketing and product decisions.
- Foundation for future AI and analytics projects.

The system can help management understand customer opinions without manually reading thousands of reviews.

8. Conclusion

This project demonstrates the feasibility of using entry-level machine learning techniques for automated sentiment analysis. Using TF-IDF and Logistic Regression, customer reviews can be classified efficiently with minimal computational resources.

The prototype serves as a proof-of-concept for TrendCart and can be expanded in the future with advanced natural language processing techniques, larger datasets, and real-time analytics capabilities.